

PET and Gestational Hypertension 07: Medications and Regimens with Mechanisms

Companion reading handout for simultaneous listening.

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This episode is about drug choices, but we will not learn them as a shopping list.

In hypertensive disease of pregnancy, every medication has a job. Some prevent stroke. Some prevent seizure. Some stabilize BP over days. Some protect the fetus from prematurity complications. Some are safe postpartum because the fetus has delivered. If you know the job, the regimen becomes easier to remember.

Acute Severe Hypertension

Start with acute severe hypertension.

The job is stroke prevention. Severe BP means systolic at least one hundred sixty, diastolic at least one hundred ten, or both, persistent long enough to be real. The goal is not normal blood pressure. The goal is to leave the severe range without crashing uteroplacental perfusion.

ACOG first-line acute options include intravenous labetalol, intravenous hydralazine, and oral immediate-release nifedipine.

Labetalol blocks alpha one receptors and beta receptors. The alpha effect lowers vascular resistance. The beta effect blunts heart rate and contractility. ACOG dosing begins with ten to twenty milligrams IV, then twenty to eighty milligrams every ten to thirty minutes, up to a maximum cumulative dose of three hundred milligrams, or an infusion of one to two milligrams per minute.

Avoid or be cautious with labetalol in asthma, heart block, bradycardia, or decompensated cardiac disease. The mechanism explains the cautions. Beta blockade can worsen bronchospasm and bradycardia.

Hydralazine directly relaxes arteriolar smooth muscle. ACOG dosing is five to ten milligrams IV or IM, then five to ten milligrams every twenty to forty minutes, up to twenty milligrams, or infusion zero point five to ten milligrams per hour. It can cause headache, flushing, tachycardia, and hypotension.

Nifedipine blocks L-type calcium channels in vascular smooth muscle. Immediate-release nifedipine is useful when IV access is delayed or not available. ACOG dosing is ten to twenty milligrams by mouth, repeat in twenty minutes if needed, then ten to

twenty milligrams every two to six hours, maximum one hundred eighty milligrams per day. Do not use sublingual nifedipine.

The Hadassah delivery-room crisis protocol is similar: labetalol ten to twenty milligrams IV, then twenty to eighty milligrams every twenty to thirty minutes, maximum three hundred milligrams; hydralazine five milligrams IV or IM, then five to ten milligrams every twenty to forty minutes; nifedipine ten to twenty milligrams by mouth, repeat within thirty minutes if needed.

Stable or Nonurgent BP Treatment

Now stable or nonurgent BP treatment.

Here the guidelines diverge. ACOG clearly mandates urgent treatment for severe BP. NICE treats sustained BP above one hundred forty over ninety and aims for one hundred thirty five over eighty five or less. ISSHP recommends treating hypertension in pregnancy regardless of the hypertensive disorder, targeting diastolic BP of eighty five, influenced by the CHIPS trial. SOMANZ also supports antihypertensives for stable chronic hypertension, gestational hypertension, and stable non-severe pre-eclampsia. Hadassah local protocol does not use chronic medication for gestational hypertension and does not treat mild-range BP during labor.

So the resident must know both the shared rule and the local rule.

Shared rule: severe BP is urgent everywhere.

Local rule: mild-to-moderate BP treatment depends on the guideline and department.

Maintenance options include labetalol, nifedipine modified release, and methyldopa. Hadassah lists labetalol two hundred to two thousand four hundred milligrams per day divided into two to four doses, nifedipine slow release thirty to one hundred twenty milligrams per day, and methyldopa zero point five to three grams per day divided into two to three doses.

Magnesium Sulfate

Now magnesium sulfate.

Magnesium's job is seizure prevention and seizure treatment. It is not an antihypertensive.

For eclampsia, magnesium prevents recurrent seizures. For pre-eclampsia with severe features, it reduces the risk of eclampsia. ACOG emphasizes the Magpie trial, where magnesium reduced eclampsia by more than half.

Regimens vary. A common regimen is a four to six gram IV loading dose, followed by one to two grams per hour. SOMANZ recommends four grams IV loading, followed by

one gram per hour for twenty four hours total or from the last seizure. NICE critical-care guidance uses four grams IV over five to fifteen minutes, then one gram per hour for twenty four hours, with two to four grams for recurrent seizures. Hadassah uses four grams in one hundred milliliters saline over twenty minutes, then ten grams in five hundred milliliters at two grams per hour.

Toxicity is where pharmacology becomes bedside safety.

Magnesium is renally cleared. If urine output falls or creatinine rises, magnesium accumulates. Reflex loss appears before respiratory arrest. ACOG lists therapeutic levels around five to nine milligrams per deciliter, loss of deep tendon reflexes above about nine, respiratory paralysis above about twelve, and cardiac arrest at much higher levels. Hadassah uses a similar practical table: therapeutic four point eight to eight point five, loss of patellar reflex around eight point five, respiratory depression above twelve.

So monitor reflexes, respiratory rate, mental status, urine output, and renal function. If toxicity appears, stop magnesium and give calcium gluconate. ACOG uses ten milliliters of ten percent calcium gluconate IV over about three minutes.

Corticosteroids

Now corticosteroids.

Their job is neonatal, not maternal cure. If delivery before thirty four weeks is likely and the situation allows time, antenatal corticosteroids reduce neonatal respiratory distress, intraventricular hemorrhage, and other prematurity complications. A repeat course may be considered if risk persists after the initial course according to local protocol.

At thirty four to thirty six plus six, late-preterm steroids may be considered in selected patients. But ACOG is clear: if severe pre-eclampsia at or beyond thirty four weeks requires delivery, do not delay delivery solely to complete steroids.

Magnesium can also be used for fetal neuroprotection when early preterm delivery is imminent. That is a separate indication from maternal seizure prophylaxis. Same drug, different job.

Postpartum Medication

Now postpartum medication.

After delivery, there is no fetus exposed through the placenta. That changes the medication landscape. NICE and ISSHP allow breastfeeding-compatible agents such as labetalol, nifedipine, hydralazine, and in many cases ACE inhibitors such as

enalapril or captopril. Methyldopa is often avoided postpartum when alternatives exist because of mood concerns, though data are limited.

NICE recommends starting postpartum treatment if a patient not previously on medication has BP at least one hundred fifty over one hundred. ISSHP targets diastolic eighty five postpartum, similar to antepartum.

NICE also gives practical postpartum details. Methyldopa used in pregnancy should be stopped within two days after birth. Enalapril can be used postpartum, but renal function and potassium should be checked. If one agent is not enough, combinations such as nifedipine and enalapril are reasonable in that framework, with labetalol or atenolol added or substituted when needed. Diuretics and angiotensin receptor blockers are generally avoided for hypertension treatment in breastfeeding women when possible.

NSAIDs

Finally, NSAIDs.

This is a guideline-tension medication. ACOG reviews reassuring evidence that ibuprofen and other NSAIDs do not meaningfully worsen postpartum BP in pre-eclampsia with severe features. ISSHP allows NSAIDs if other analgesics are ineffective and there is no acute kidney injury or risk factor for it. SOMANZ is more cautious and recommends against routine NSAID use in pre-eclampsia until more safety data are available, while allowing short inpatient use when alternatives are inadequate.

The practical rule is not never and not always. Avoid NSAIDs or monitor closely in acute kidney injury, chronic kidney disease, uncontrolled BP, sepsis, hemorrhage, or fluid overload risk.

Closing Synthesis

Let us close by sorting the medications by job.

Labetalol, hydralazine, and nifedipine treat dangerous BP. Magnesium prevents seizures. Corticosteroids help the preterm fetus. Maintenance antihypertensives prevent severe BP recurrence. Postpartum medications protect the mother when fetal exposure is no longer the same issue.

Resident Reflex

The resident reflex is this: before choosing the drug, name the danger. Stroke, seizure, prematurity, recurrent hypertension, pulmonary edema, or postpartum vascular risk. The right regimen is easier when the job is clear.